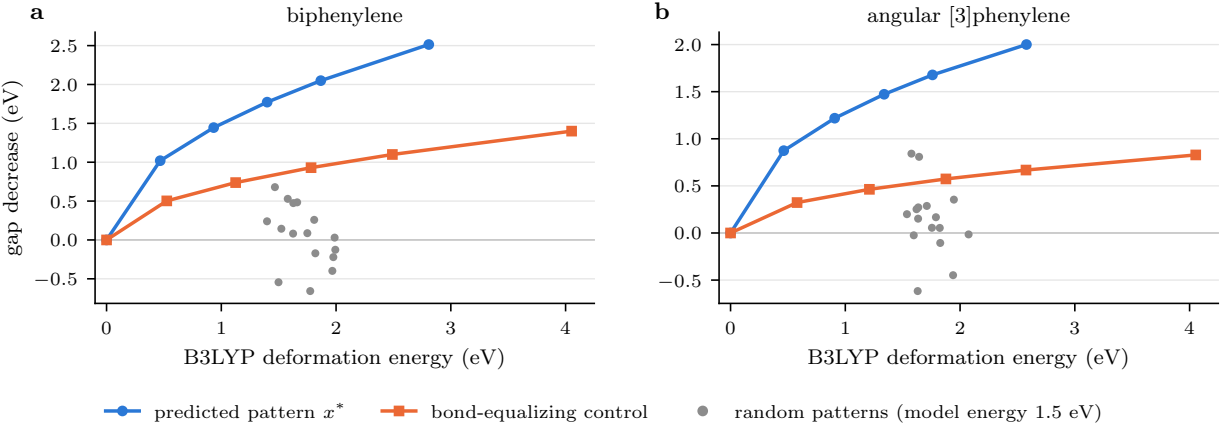

a
biphenylene
gap decrease (eV)
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
B3LYP deformation energy (eV)
b
angular [3]phenylene
2.0
1.5
1.0
0.5
0.0
−0.5
0
1
2
3
4
B3LYP deformation energy (eV)
predicted pattern x^*
bond-equalizing control
random patterns (model energy 1.5 eV)